

CMN02A1

ASSIGNMENT #2

DUE DATE: 23 MAY 2025 [BEFORE MIDNIGHT]



Instructions

Complete the following questions and submit your typed-out solution before the submission deadline, which is **Friday, 23 May 2025** at **23:59:00**, via Moodle. Please note the following instructions:

1. This is the second individual assignment (Assignment 2) for CMN02A1, 2025
2. This assessment counts **30%** towards the **Final Mark** for the module
3. The general University of Johannesburg policies, procedures and rules pertaining to written assessments apply to this assessment
4. Strictly, **no extensions will be granted** for whatever reason
5. Late submissions will be assessed and will carry a strict penalty of **5%** for each day (excluding public or university holidays) that the assessment is late for a maximum of 3 days (this means that there will be no submission possible after **26 May 2025, at 23:59**)
6. Read through the assessment to make sure that you understand the requirements
7. **IMPORTANT:** Please name your submission like so (failure to do so will attract a **5%** penalty):
 - [STUDENT_NO]_[SURNAME]_ASS2.pdf
 - (e.g. 2018993320_Brown_ASS2.pdf)
8. **PLAGIARISM:** Please refer to your Learning Guide as well as the latest University of Johannesburg's plagiarism policy document entitled: "POLICY: PLAGIARISM"
9. **COPYING:** This is an individual assignment; if any copying is detected, all parties involved will score a **0%** for the assignment and **WILL** face disciplinary consequences

GOOD LUCK!

SECTION A — CALCULATIONS AND CHARTING

[TOTAL: 23]

Question 1 (10)

A system is going to transmit the byte **0010 0100**.

1.1 Show the four check bits **c8**, **c4**, **c2** and **c1** that will be added to this byte. **SHOW ALL YOUR WORKING.** (8)

1.2 Show the final bit string that will be transmitted. (2)

Question 2 (8)

Consider the IPv4 address **132.155.121.2/25**

2.1 Within which IP class does this address fall? (2)

2.2 How many addresses are in this IP address block? (2)

2.3 What is the network mask in dotted binary notation? (2)

2.4 How many bits belong to the host ID? (2)

Question 3 (5)

In a sliding window error control system, **Station A** sends a packet with bytes 201 – 400, followed immediately by a packet with bytes 401 – 600. Create a diagram of this error control scenario and show the response(s) that **Station B** will send if the second packet is lost in the network.

SECTION B — DEFINITIONS AND SHORT ANSWERS

[47]

Question 4 (8)

What are some of the error prevention techniques that can be adopted to guard against the following types of errors?

4.1 Impulse noise (2)

4.2 Crosstalk (2)

4.3 Echo (2)

4.4 Jitter (2)

Question 5 (6)

During data transmission, there are three (3) actions that a receiver can take if it detects an error in a data packet/frame. **NAME** and **BRIEFLY DESCRIBE** these actions.

Question 6 (6)

Consider the IEEE standard **802.3ad-2000**.

6.1 What does this standard describe? (2)

6.2 Why would a network provider want to implement this standard? **GIVE FOUR (4) REASONS.** (4)

Question 7 (11)

Distinguish between each of the following:

7.1 Error-detection, error-control, and error correction (3)

7.2 Station and node (2)

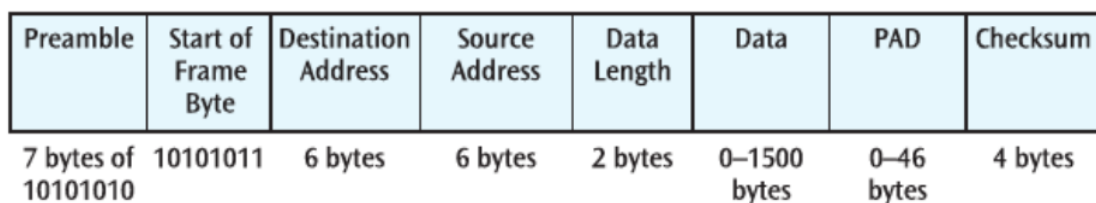
7.3 Hop limit and hop count (2)

7.4 DHCP and NAT (2)

7.5 Shared segment network and dedicated segment network (2)

Question 8 (10)

Consider the diagram below:



8.1 What frame format is this diagram for? (2)

8.2 What is the minimum size that any station can transmit for this type of frame? (2)

8.3 What are the **Destination Address** and **Source Address** fields for? (2)

8.4 Why are the **Destination Address** and **Source Address** fields 6 bytes long? (2)

8.5 What is the main application or use of the **Checksum** field? (2)

Question 9 (6)

The **virtual circuit packet-switched network** was created in order to primarily address some shortcomings inherent in the datagram packet-switched network.

- 9.1 What are the main characteristics of a virtual circuit packet-switched network? (3)
- 9.2 What are two of its advantages? (2)
- 9.3 What is its main disadvantages? (1)

Mark Allocation

Section	Marks
Section A – Calculations and Charting	23
Section B – Definitions and Short Answers	47
TOTAL	70

Table 1: Assignment 2 Mark Allocation

== ENDS ==